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MILITARY SPECIFICATION

ALIGNMENT DEVICE, OPTICAL, GUN BORE

Reactivated after February 23, 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the M139, M140, M139A1, M140A1 and M154 Alignment Devices which provide a close-in optical reference to check the boresight of the fire control at the weapon site. They each have an illuminated reticle.

1.2 Classification. The Alignment Device is used to provide an on-carriage optical reference to periodically check the boresight setting of the fire control and there are the following types:

1.2.1 M139. This Alignment Device is intended to support the M198 Howitzer. and uses tritium.

1.2.2 M139A1. This Alignment Device is intended to support the M198 Howitzer and does not use tritium.

1.2.3 M140. This Alignment Device is intended to support the M119A1 Howitzer and M109 Howitzer Series and uses tritium.

1.2.4 M140A1. This Alignment Device is intended to support the M119A2 Howitzer and M109 Howitzer Series and does not use tritium.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-AIS-SS, Picatinny, New Jersey 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

AMSC N/A FSC 4931

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1.2.5 M154. This Alignment Device is intended to support the M777 Howitzer Series and does not use tritium.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not documents are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-13830	-	Optical Components of Fire Control Instruments; General Specifications Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	-	Fire Control Materiel, Manufacture, and Inspection, General Specification for
MIL-I-45607	-	Inspection of Equipment, Supply and Maintenance of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of federal and military specifications, standards, and handbooks are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

F11741648	-	Alignment Device, Optical Gun Bore
F8565556	-	Pressure Tester, Two Station
F9333945	-	Fixture, Final for Alignment Device M139/M140

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DOCUMENTS

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 9000 - Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to ANSI New York City Office Operations, 25 West 43rd Street, 4 floor, New York, NY 10036 or go online to http://www.ansi.org)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DSAM 4145.8 - Radioactive Commodities in the DOD Supply System

(Application for copies should be addressed to the Superintendent of documents, Government Printing Office, Washington, DC 20402.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Regulations pertaining to radioactive devices. Marking, identification, handling and shipping of radioactive devices shall be in accordance with DSAM 4145.8 and MIL-STD-129 as applicable.

3.3 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

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3.4 Fabrication. The Alignment Device shall be manufactured in accordance with Drawing F11741648 and drawings pertaining thereto.

3.5 General specifications. The contractor shall be responsible for compliance with the requirements of specification MIL-F-13926.

3.6 Orientation. Whenever specified herein, the conditions listed below shall apply:

a. Dovetail base surface in a horizontal plane.
b. Side of dovetail parallel to a 4 power optical reference telescope.
c. Alignment Device latched against side of dovetail.

3.7 Temperature conditions. Except as specified in 3.8.1, 3.8.2. 3.9.1; the requirements of this specification shall be met at a standard ambient temperature between 60 º F and 90 ºF (16 ºC to 32ºC).

3.8 Environmental service conditions.

3.8.1 Storage temperature. The Alignment Device shall show no evidence of physical failure when thermally stabilized at ambient temperatures of +160 ºF (+71ºC) ± 5°F (±3 °C) and -60 °F (-51 °C) ± 5 °F (± 3 °C) for a period of 4 hours ± 15 minutes at each temperature.

3.8.2 Operating temperature. . The Alignment Device shall meet the applicable requirements of 3.5.4.2 while exposed and thermally stabilized at ambient temperatures of +145 °F (63 °C) ± 5°F (±3°C) and -50 °F (-46°C) ± 5°F (± 3°C) for a period of 4 hours ± 15 minutes at each temperature. Upon return to standard ambient temperature, +60 °F to +90°F (+16°C to + 32°C), from each operating temperature, the Alignment Device shall meet the requirements of this specification.

3.8.3 Shock. The Alignment Device in its carrying case (see 6.9) shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axis. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the Alignment Device shall not have been damaged or adversely affected and shall meet the requirements of 3.8.5 and 3.9 (see 4.6.4).

3.8.4 Vibration.

3.8.4.1 Vibration "A". The Alignment Device in its carrying case (see 6.9) shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibrations in each direction shall be at frequencies varied logarithmically from 5 to 500 Hz. The double amplitude of applied vibrations shall be .036 inches at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the Alignment Device shall not have been damaged or adversely affected and shall meet the requirements of 3.8.5 and 3.9 (see 4.6.5.1).

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3.8.4.2 Vibration "B". With the Alignment Device held by the upright tube in a horizontal position, the Alignment Device shall be vibrated in a vertical plane at a constant frequency of thirty Hz with amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus 15 seconds. Subsequent to vibration, the Alignment Device shall not have been damaged or adversely affected and shall meet the requirements of 3.8.5 and 3.9 (see 4.6.5.2).

3.8.5 Effects of vibration and shock. This requirement shall be met with conditions of 3.6 reestablished. The line of sight established prior to vibration or shock shall not deviate more than .1 mil in azimuth due to subjecting the Alignment Device to Vibration "A", Vibration "B", or Shock (see 4.6.6).

3.9 Performance.

3.9.1 Sealing. The Alignment Device shall be capable of sustaining an internal pressure of 10 pounds per square inch gage (psig) for a period of 2 hours ± 10 minutes at a temperature of 75 ° F ± 15 °F. At the end of the 2 hour period, the leakage shall not have exceeded 0.2 psig. The pressure within the Alignment Device shall then be returned to 1 to 2 pounds per square inch gage (psig) and the optical cavity sealed. The interior atmosphere of the optical cavity of the accepted Alignment Device assembly shall be nitrogen in accordance with 3.9.2 (see 4.6.1).

3.9.2 Purging. The interior of the Alignment Device shall be flushed with dry nitrogen until the gaseous mixture bled from the Alignment Device has a dew point of at least as low as -25 ° F (see 4.6.8).

3.9.3 Cleanliness.

3.9.3.1 Dirt. Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle surface shall not exceed the width of the reticle line. The maximum number of dirt particle appearing within a central field of view of 50 mils diameter shall be three (3). The maximum number of dirt particles appearing outside this field of view shall be three (3). The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than ¼ of the reticle line width shall be ignored regardless of distribution.

3.9.3.2 Defects. There shall be no evidence of moisture, finger prints, condensates, fractures, and adhesive separations on any glass component. Internal inspection shall be made by viewing through the objective end of the Alignment Device (see 4.6.9).

3.9.4 Accuracy.

3.9.4.1 Collimation. With conditions of 3.6, the line of sight of the device reticle shall be parallel in elevation with the base of the dovetail within 0.1 mil and parallel in azimuth within 0.1 mil to the side (angular surface) of the dovetail (see 4.6.10).

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3.9.4.2 Replacement repeatability. When the Alignment Device is removed from the reference dovetail mount and subsequently remounted, the device reticle line of sight shall meet the requirements of 3.9.4.1 and shall not have changed from the original position by more than 0.1 mil (see 4.6.10).

3.9.5 Optical characteristics.

3.9.5.1 Parallax. Using a collimator with the reticle adjusted to be parallax free for a 80 meter target, the parallax between the reticle and the cross on the Alignment Device reticle shall not exceed 0.1 mil (see 4.6.11).

3.9.6 Operability of moving parts.

3.9.6.1 Attachment. The Alignment Device shall meet all functional requirements of this specification when the Alignment Device mounting bracket or a suitable substitute is employed (see 4.6.10).

3.9.6.2 Latch. The latch shall be free to rotate to lock the Alignment Device to the dovetail on the Alignment Device's mounting bracket and shall be free to rotate to release the boresight from the dovetail. This requirement shall be met at standard ambient conditions and at the temperatures specified in 3.8.2 (see 4.6.10).

3.9.7 Illumination.

3.9.7.1 Reticle. The reticle marking shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into night through a 4-power telescope at a distance of 10 feet ± 1 foot. This observation must be made with and without the use of a 50% neutral density filter (see 4.6.12).

3.9.8 External tritium contamination. Any part of tritium shall not be present on the external surfaces of the M139 and M140 alignment device in excess of 1000 dpm/100 sq. cm. Subsequent to the requirements of 3.8.1 through 3.8.4.2 (see 4.6.13) as required by paragraph 4.3.2.2 Final acceptance inspection provisions, or paragraph 4.4 Special sampling provisions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in ISO 9000.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) Alignment Devices plus three (3) each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production (see 4.4.2).

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-1916.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-1916 and the sampling plans specified in Tables I, II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I – 100% Inspection

NO	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101	Orientation	3.6	4.6.1
102	Vibration “B”	3.8.4.2	4.6.5.2
103	Effects of Vibration & Shock	3.8.5	4.6.6
104	Sealing	3.9.1	4.6.7
105	Purging	3.9.2	4.6.8
106	Cleanliness	3.9.3	4.6.9
107	Reticle Accuracy (Collimation)	3.9.4.1	4.6.10
108	Repeatability	3.9.4.2	4.6.10
109	Parallax	3.9.5.1	4.6.11
110	Operability	3.9.6	4.6.10
111	Reticle Illumination	3.9.7.1	4.6.12
112	Radiological Contamination	3.9.8	4.6.13

NOTE: The tests in Table I shall be conducted at a temperature between +60°F and +90°F.

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-1916.

4.4 Special sampling.

4.4.1 General. Three Alignment Devices shall be selected at random by a Government representative as a special sampling from each 50 produced. The samples shall meet the requirements and tests in Table II and shall then meet the requirements and tests in Table I.

TABLE II. Special sampling

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
201	Fabrication	3.4	Applicable drawings – visual
202	General Specification	3.5	MIL-F-13926/Visual
203	Storage Temperature	3.8.1	4.6.2
204	Operating Temperature	3.8.2	4.6.3

4.4.2 First article testing. Inspection shall be performed on the following characteristics during First Article Testing (see 4.2). Prior to performing these tests, the selected items shall have met the inspection criteria outlined in Tables I and II. Upon

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successfully meeting the inspection criteria outlined in Table III, the selected items shall be re-inspected for those characteristics of Table I.

TABLE III. – First article testing

NO	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401	Shock	3.8.3	4.6.4
402	Vibration “A”	3.8.4.1	4.6.5.1

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment comparable with the “Test Methods and Procedures” specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of the test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. In conjunction with 4.5.2.2, the following Standard Test Equipment shall be utilized in the performance of the applicable tests specified in 4.6.

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	Nomenclature	Description
1.	Temperature chamber; Vibration Equipment; and Shock Equipment	Standard type conforming to the accuracy outlined under "Test facilities" in MIL-F-13926

4.6 Test methods and procedures.

4.6.1 Orientation. Test fixture conforming to Drawing F9333945 will be used for the inspection of the Alignment Device, Optical, Gun Bore. Affix the test fixture on a leveled, vibration free surface in accordance with the specified set-up instructions. The calibration and set-up instructions are the prime requirements necessary to establish the conditions of 3.6, prior to positioning the Alignment Device under test to the test fixture. The Alignment Device under test shall be oriented as specified in 3.6. Operating instructions for final inspection fixture are detailed on Drawing F9333945.

4.6.2 Storage temperature. The Alignment Device shall be exposed to each temperature extreme specified in 3.8.1 for a minimum of 4 hours. After each temperature exposure, the Alignment Device shall be stabilized at standard ambient conditions and subjected to the examinations and tests to determine conformance with the requirements of 3.8.1. Caution shall be exercised during this test to avoid subjecting the Alignment Device to thermal shock.

4.6.3 Extreme operating temperature. The Alignment Device shall be exposed to each temperature extreme specified in 3.8.2 for a minimum of 4 hours. After exposure for 4 hours and while at the test temperature, operability of latch shall be in accordance with 3.9.6.2. After each temperature exposure, the Alignment Device shall be stabilized at standard ambient conditions for a minimum of 4 hours and subjected to the examinations and tests to determine conformance with the requirements of 3.8.2. Caution shall be exercised during this test to avoid subjecting the Alignment Device to thermal shock.

4.6.4 Shock. The Alignment Device in its carrying case shall be tested in accordance with 3.8.3. Subsequent to testing, the unit shall be subjected to the examinations and tests to determine conformance with the requirements of 3.8.3.

4.6.5 Vibration.

4.6.5.1 Vibration "A". The Alignment Device while in its carrying case shall be vibrated in accordance with 3.8.4.1. Upon completion of the vibration cycling, the Alignment Device shall be subjected to the examinations and tests to determine conformance with the requirements of 3.8.4.1.

4.6.5.2 Vibration "B". The Alignment Device shall be secured to the vibration table by suitable adapters with the tube and head parallel to the vibration table surface. Lever, 10544452, shall be immobilized. The Alignment Device shall then be vibrated in accordance with the conditions specified in 3.8.4.2. Upon completion, the unit shall be

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subjected to the examinations and tests to determine conformance with the requirements of 3.8.4.2.

4.6.6 Effects of vibration and shock. Subsequent to vibration or shock, the Alignment Device shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.8.5, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts. An audible examination shall be made of the unit by shaking it and listening for loose internal parts or broken optical components, such as lenses and reticle, etc. Conditions of 4.6.1 shall be reestablished and the line of sight established prior to vibration or shock shall not have deviated in excess of the tolerance specified in 3.8.5. Any one of the possible defects defined above shall be cause to consider an Alignment Device defective. Items which have passed the above examinations shall be subjected to the tests specified in 4.6.7 through 4.6.13.

4.6.7 Sealing. The internal pressurizing of the Alignment Device to test sealing shall be performed with special testing equipment conforming to Drawing F8565556, Pressure Tester-Two Station. The pressure shall be applied through an appropriate adapter. Pressurize the unit with dry nitrogen gas to 10.0 ± 0.10 psig. The dry nitrogen gas shall have a dew point at least as low as minus 25 °F. The applied pressure shall be retained for a minimum of 2 hours. Any evidence of leakage at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.9.1, as measured on the pressure gage of the testing equipment.

4.6.8 Purging. Upon satisfactory completion of the sealing test, the Alignment Device shall be flushed with dry nitrogen gas. The flushing shall be accomplished by opening the purging vent and allowing the nitrogen gas to flow through the Alignment Device until the bled gas conforms to the dew point requirements of 3.9.2 as measured with a dew point tester. Close the purging vent.

4.6.9 Cleanliness. The Alignment Device, with Collimator Assembly D7573433, shall be subjected to the applicable test procedures outlined in MIL-PRF-13830 to determine compliance with the requirements of 3.9.3.1 and 3.9.3.2.

4.6.10 Reticle accuracy, (collimation). With the Alignment Device positioned as specified in 4.6.1, observe where the Alignment Device's boresight cross falls upon the target in the Collimator Assembly. The intersection of the boresight cross shall not deviate from the collimator target zone in excess of the limit specified in 3.9.4.1. Remove and reposition the unit under test three times. At each resetting into the fixture, the boresight cross in the Alignment Device shall remain centered within the specified limits to ascertain compliance with requirement of 3.9.4.2.

4.6.11 Parallax. Test shall be performed using the set-up in 4.6.1 with the 80 meter focus in the Collimator Assembly. Parallax shall be recognized as any apparent displacement, when the observer's head is moved from side to side or up and down. Any

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movement between the collimator's reticle and the Alignment Device reticle shall not exceed the value specified in 3.9.5.1. (The width of a reticle line in the collimator is equal to 0.10 mil).

4.6.12 Reticle Illumination. The test shall be performed in an area in which ambient lighting can be varied. Personnel performing the testing are to be dark adapted by being in the darkened room for at least 15 minutes before viewing the reticle. The reticle shall be viewed through the collimator assembly using a 4-power telescope at a distance of 10 feet ± 1foot. The reticle marking shall be clearly distinguishable with and without the use of a 50% neutral density filter while the light level is varied gradually from 0 to 10 foot candles over a period of 1 to 2 minutes.

4.6.13 Radiological contamination. Contamination test shall be performed by wiping the M139 and M140 Alignment Device with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The scintillating system used to measure contamination must be calibrated. The readout shall be within 10% of the known standard value when counted to a total of 2000 disintegrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.9.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The Alignment Device is used to provide an on-carriage optical reference to periodically check the boresight setting of the fire control.

6.2 Ordering data. Procurement should specify the following:

a. Title, date and number of this specification.
b. Selection of an applicable level of preservation, packing and packaging.
c. Applicable stock number.
d. Provisions of First Article testing.

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6.3 First article. When a first article is required, it must be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article sample should consist of the first production items and as specified in 4.2. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

6.4 Radioactive material. The M139 and M140 Alignment Devices contain tritium, a radioactive material. Prospective bidders should be cognizant of the need for an Atomic Energy Commission License governing the procurement and handling of radioactive material.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Packaging data sheet drawings. The following packaging data sheet drawings should be obtained from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIL-P B455, Picatinny, NJ 07806-5000.

SPI AM11741648 - Special Packaging Instruction for the M139 Alignment Device

6.7 Hazardous material.

6.7.1 Tritium. The M139 and M140 Alignment Devices contain tritium, a radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, processing, handling and shipping of radioactive material.

6.7.2 Lithium Batteries. The M139A1, M140A1 and M154 utilize Lithium batteries. Prospective Bidders must be cognizant of the need to handle and/or dispose of these batteries in accordance with all Local, State and Federal Laws.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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6.9 Carrying case. The applicable carrying case will be provided as Government Furbished Equipment (GFE) by the procuring agency.

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 4931-2006-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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Check the source to verify that this is the current version before use.